Abstract

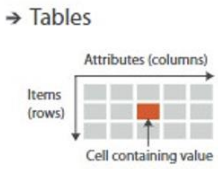
The project deals with datasets from the 2011 census in England and Wales. The aim is to design, construct, and evaluate an exploratory data analysis of census data using suitable visualisation techniques and data projection. The study strives to explore the aspect of employability and socio-economic life in England and Wales and their relationship with industry and the public's qualifications. This analysis clarifies various tasks related to this topic and presents it to the end-user in the most effective way to perceive it according to the vis principles.

Introduction

The study analyses the economic activities in England and Wales and their relationship with the industry and the public's qualifications.
It is necessary to explore these variables and how they relate to each other to get an overall picture of the economic status in England and Wales.
In this study, census data in England and Wales of 2011 is explored, which revolve around three aspects: economic activity, qualifications and Industry.
Performing an exploratory data analysis on these datasets can help to gain useful insights to help in making better-informed decisions.
The economic activity dataset represents information that classifies usual residents by economic activity and additional information about the unemployed population (including full-time students), with estimates of the number of long-term unemployed, those who have never worked, and the classification of unemployed by key age groups. The qualification dataset outlines the information that classifies usual residents by their highest level of qualification and the industry dataset describe information that classifies usual residents by the industry in which they work.
The study aims to analyze and examine this data to discover useful findings and present it smoothly and efficiently for end-users to interact with it.
The outline approach is to design, create, and evaluate an exploratory analysis of the census dataset using different visualisation techniques and data projection.
The rest of the report is organized as follows: Data Preparation and Abstraction is presented in Section 2. In Section 3, I describe the Task Definition. The visualisation Justification that was based on to derive and present the plots are described in Section 4. Evaluation of the work is described in Section 5, conclusions in Section 6.

Data Preparation and Abstraction

The data has been collected from the National Offender Management Information System (NOMIS) website using its query search, which provides various selection of tables for census 2011.



Dataset Structure

The dataset consists of various static tables. The variables/attributes types are nominal such as country and gender, ordinal such as qualification levels and discrete measures.

In total, three main tables were queried and downloaded based on specific features. All datasets have the "county" dimension as the primary key to link all of them. There is also a map file called "UKTableauFilledMAP" that helps to plot the UK filled map in tableau.
I applied several preprocessing steps to get the data ready for the analysis.

- **County names replacement:** Some county names in the datasets does not match the exact name in the map file hence lead to issues in plotting the geographic data later. For example, “City of London" was written in the column of most data tables and needed be replaced with “Greater London" to match the same name in the map file.
- **Pivoting:** Some dimensions needed to be pivoted in tableau to make it more organised and easier to use for the visualisations. For instance, “Male" and "Female" dimensions needed to be pivoted in all the rows.
- **Removal of unused columns**: Some dimensions in the original dataset were removed as they can be done with simple calculations in tableau. For example, “All Categories" dimension in the industry table needed to be removed as I can sum all the dimensions values in tableau hence it was an extra dimension in the dataset.
- Finally, I rearranged columns, aggregated related data as a hierarchy in tableau and changed the data types for each dimension to be the correct one. For example, setting the geographic role of the "County" dimension to be county and "Country" to be Country/Region role.
After preprocessing the datasets are represented on average without unused data for the analysis.

Task Definition

The main task is to enable the user to investigate the distribution of employment and unemployment rates in England and Wales and examining if there is gender bias on the socio-economic life as well.

This can be done through the following sub-tasks questions:

- Is there any imbalance of male and female employment/unemployment rates in England and Wales?
- Which region has the highest employment rates?
- Are employed/unemployed males and females proportionally distributed and have similar rates across all counties in England and Wales? Is there any gender bias in a particular county?
- What are the figures of the types of employment for males and females in England and wales?
- Do the counties that have higher numbers of economically active residents tend to have more/less economically inactive residents?
- Which qualification does the majority of residents in England and Wales hold?
- What qualifications do residents hold in each county? are there any differences?
- What are the top counties in terms of having higher numbers of retired residents?
- Prominent Industry in England and Wales employing the largest number of residents?
- Is there a fair distribution of industry economic activity in England and Wales?

The descriptions of the visualisations are as follows. Representing various map visualisations to show employment and unemployment rates in England and Wales and specifically in each county

as it is the level of granularity of the plots. The distribution of employment and unemployment are represented through these map visualisations to give a clear overview of the economic status in each county. It can be seen from the maps which region/county has the highest employment rates and whether males and females proportionally divided and have similar rates across all counties in England and Wales. Similarly, a diverging bar chart to display the figures of unemployability for both genders in each county in England and wales. This visualisation aims to show which counties suffer from the unemployment issue. For example, Birmingham has the highest numbers of unemployment rates for both genders. Additionally, a scatter plot visualisation to explain the relationship between the numbers of economically active residents and inactive residents in both regions. Moreover, a horizontal bar chart presents the distribution of the employment types for both males and females. One finding that could be found from this visualisation task is that females tend to have Part-time jobs more than males. Plus, an interactive bar graph to show qualifications distribution in each county. Another visualisation is the bubble chart that reveals the top counties in terms of having higher figures of retired residents. Finally, a grouped bar chart to display various industries employment rates. The task of this visualisation is to see the employment distribution of various industries in England and Wales.

Visualisation Justification

Here are the reasons for the visualisations techniques I have chosen to perform each task according to the principles of info vis.

- **Employment Residents per County Visualsation Technique**

  Here we would like to show the user the distribution of employment rates in each county in England and Wales. I have chosen to represent this task in an interactive map to give the user a general overview of the data distribution. This visualisation technique was based on the role "Overview first, zoom and filter, details on-demand" which is one of Munzner's 8 Rules of Thumb. Here the user can have a general observation of the data distribution on the map. The overview is meant to give the user a broad knowledge of the entire information space. Hence, the user does not need to navigate the map or scroll to find the regions that have the highest/lowest employment rates. The user can also investigate further the details of the employment figures in each county by zooming in or hovering on each county. Additionally, the user can filter the regions shown on the map by using the Country filter next to the map figure and there is also a measure values indicator which shows that the darker the colour on the map, the larger the number of employed residents in each county. Plus, the colours tend to have the same luminance to ease the effort on the user's eyes.

- **Unemployed Male and Female Visualisation Technique**

  In this task, I chose to visualise the data in a diverged bar chart to represent a clear comparison between male and female's unemployment rates in each county. I chose this technique to visualise the data of this task because the best vis designs should reflect in terms of both form and function. It is one of the vis design principles that aim to be both beautiful and effective in presenting the information.

- **Types of Employment Visualisation Technique**

In this task, I chose to represent the types of employment for both genders in a way that is easy and eye-catching for the user to recognise the information without memory dependence. Since using the user eyes to see presented information in the graph has a much lower cognitive load than consulting our memory to memories details, A horizontal bar chart is a great visualisation technique in this case. Also, I made distinct colours that signify each gender (blue for male and pink for female) along with male and female icons on each bar to make it more convenient for the user to recognise.

- **Top Counties that have Retired Residents Visualisation Technique**

Representing the top counties that have the most retired resident's figures in a bubble chart is effective because it makes seeing the numbers more clearly for the user through the size of the bubble and the depth of the colour. It encodes channels to be most noticeable.

- **Relationship between Economic active and inactive residents Visualisation Technique**

A Scatter plot is the suitable visualisation method to compare between the two numerical dimensions (active and inactive residents) in England and Wales and to check what kind of relationship they have. The user can see the positive correlation between the two variables as there is a reference line to help guide the user to see if there are any outliers. Also, it helps to support the pattern-finding capability of the brain by turning to turn information structures into patterns on the scatter plot graph.

- **Interactive Qualification Distribution Visualisation Technique**

In this task, I intended to plot the data interactively and responsively to the user. I used a vertical bar chart to show the overall distribution of resident's qualifications in England and Wales. I also put a handy filter containing all the counties. So, the user can choose "Bristol" for example to see the qualification distribution in this specific county. This is to apply one of the vis principles which is the interaction and how much it takes for the graph to respond to user action. The user can change the graph according to the county they search for with a simple click.

- **Industry Distribution Visualisation Technique**

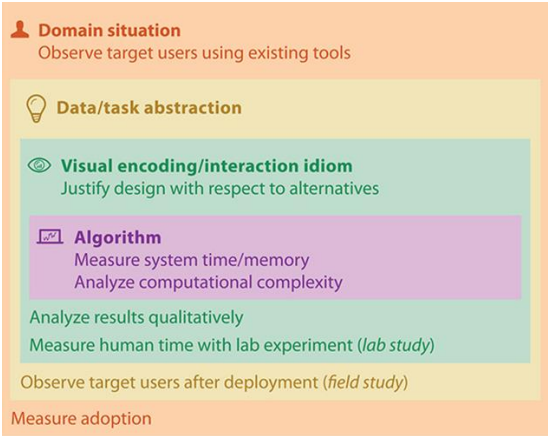
Here I try to show the user various industry types and their figures in England and Wales regions. I chose the grouped bar chart to group England and Wales for each industry type because all the items here share the same level of the categorical attribute "Country" and this can be perceived as a group. Therefore, it does not add disorder in the form of the bar chart lines.

- **Dimensionality Reduction (PCA)**

Here I Included a feature dimensional analysis for the economic activity per county. The economic activity dataset has high dimensionality features and I reduced them into low dimensionality features with two principal components. Then, I projected the data into a two-dimensional space and clustered the data points with the county as the index label. It seems to give good information with only two features and does not have any extra knowledge.

Evaluation

Validation is very critical for the visualisation design because the vis design space is large, and most designs are ineffective. Therefore, it is a continuous process to improve the vis appearance to make it more interactive and user friendly. Here I assessed my design process based on the four-level of design as shown in the graph below.



Four levels of design

In the domain situation, I validated the group of target users, their domain, their questions and their data. Users have a certain vocabulary to describe the problem domain. Hence, I addressed the tasks of my visualisations to my classmate's evaluators by writing a brief description of the main tasks and sub-tasks. Then, I asked them in the questionnaire form questions related to the domain of the study to answer them on a scale-out of 5 where 5 is the highest confidence. An example question is whether the visualisations techniques clearly illustrate the main goal of the study to the user and fully display the information that needs to be explored. Moreover, validating if these visualisations help the audience to clearly understand the result of the research.

Regarding the data/task abstraction, I seek to reduce the cognitive load for the user by making sure that the visual representation addresses the task problem. The questions I asked the users in this validation section including whether the goals of the tasks clear and well defined. Similarly, If the data for both tasks well abstracted and used, that is, the tasks are conveyed to a generic description. I also asked the user to give their feedback on the definitions of the tasks presented such as Economic activity in each county.

In the visual encoding/interaction idiom, I validated to make sure that
the design of the visualisation is guided by the abstract tasks and measure my idiom design choices. So, I asked the user in the questionnaire form to reflect on my visualisations design for good interactivity. Besides, assessing my visual graphs in giving the user a clear and

straightforward perception. The colours are another aspect here whether they signify both genders in the study clearly.

Finally, for algorithm evaluation, the goal is to measure the computational complexity of the algorithm.
I asked the user to evaluate the design of the PCA dimension reduction algorithm and the K-means clustering algorithm.

Conclusion

Overall, regarding the study task, there is a close relationship between the employment statistics of the population of England and Wales and the level of education of the public, which reflects this in the distribution of different jobs in the industry. As is evident in the analysis that I conducted in this study, there is a significant bias in employment, with most of them centred in England more than Wales and especially in the southwest region. As for the visual analysis presentation, I learned that there are many forms of displaying information, but it requires a lot of effort, thinking and experiments to choose the most appropriate visualisation technique that makes access to information by the user easier and more effective.